Palaeomagnetism of the Carboniferous E–W dyke swarm in Argyllshire

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Synopsis

The majority of 21 sampled dykes in this swarm have a coherent reversed magnetisation linked to the c. 320 Ma, Lower-Upper Carboniferous boundary, age of intrusion. Fifteen sites define a palaeopole at 354·6°E, 35·0°S (dp = 2·5, dm = 5·0), coinciding precisely with the apparent polar wander path for Europe at 320 Ma.

Geological Setting

The south west sector of the Northern Highlands is intruded by a post orogenic suite of comptonite and quartz dolerite dykes. The swarm extends from the West Coast to the Great Glen Fault where it cuts, but does not penetrate, the crush rocks of the fault zone. Dykes are abundant in the vicinities of Loch Eil and Loch Shiel but are virtually absent north of Loch Quoich. They were widely regarded as late Carboniferous to early Permian (*c.* 280–270 Ma) in age (Richey 1939, Urry and Holmes 1941, Phemister 1960), but more recent K-Ar dating (Baxter and Mitchell 1984) defines an age of 320 ± 4 Ma close to the Lower-Upper Carboniferous boundary.

Palaeomagnetic results

Between five and seven independently-oriented cores were drilled at sites in twenty one members of this swarm shown in Figure 1 with grid references included in Table 1. Of the dykes dated by Baxter and Mitchell in the Loch Eil–Loch Arkaig area sites B1 (319 Ma) and B2 (327 Ma) are close to site 20 and the five dated dykes S3 to S10 (283–334 Ma) are in the vicinity of palaeomagnetic sites 2, 3 and 4. Most samples were treated in a range of alternating fields (a.f.) up to 140 milliTesla (mT) incremented in steps of 5 or 10 mT. One core from each site was stepwise thermally demagnetised in steps of 150°C reduced to 50 and 20°C near the Curie point of magnetite (580°C) or hematite (680°C); thermal treatment above 500°C was further conducted on some of the samples after a.f. cleaning. Thermomagnetic determinations (saturation magnetisation, J_s, versus temperature) were used to determine Curie points (Table 1).

Some representative a.f. demagnetisation results are illustrated as vector plots

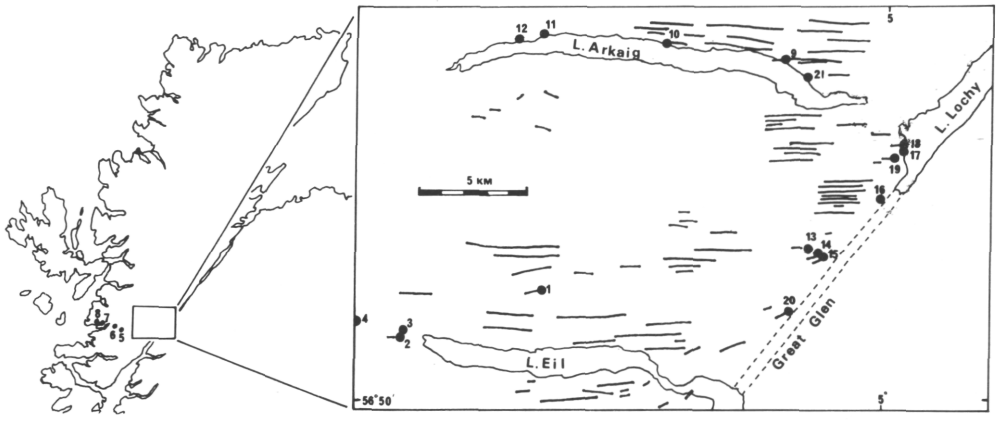


FIG. 1. Location of palaeomagnetic sampling sites in E–W dykes of the southern Northern Highlands.

(Zijderveld, 1967) in Fig. 2. The bulk of the sites fall into two groups: the majority group yields southerly negative remanence directions either initially, or with treatment in low fields. The net result of both a.f. and thermal treatment is to produce a slight reduction in inclination from shallow positive and yield cleaned directions lying close to the primitive. These remanence directions are essentially stable over the critical range of blocking temperatures although the width of these ranges is variable within the collection. At a few sites a.f. treatment has little or no effect and the remanence is unblocked close to the Curie point of hematite with a remanence direction close to the remaining sites. Some other sites illustrate a more complex magnetic structure: at 7 the vectors move through two intermediate directions before achieving a southerly negative component above 530°C retained over the critical range 530–600°C, while at 9 smeared southerly positive to negative directions converge to southerly negative components unblocked at 530°C and also removed by a.f. treatment by about 80 mT.

Sites 11 and 12 from the north shore of Loch Arkaig which may be in the same dyke (Fig. 1) have strong remanences and yield steep southerly negative directions stable to thermal treatment and unblocked between 530 and 600°C. Since these directions are indistinguishable from the shallower reversed palaeofield direction illustrated by many Tertiary dykes (and site 10 is normally-magnetised in a Tertiary direction), it seems likely that these units are part of the Tertiary swarm related to the Skye centre and are not discussed further here.

Group mean directions calculated from the linear segments of the vector plots migrating towards the origin are listed in Table 1. The predominant group of 15 sites comprises a coherent group all with southerly declinations and near-zero to shallow negative inclinations. The mean direction of D = 180·1°, I = −3·8° (α_{95} = 4·9°) corresponds to a palaeomagnetic pole at 354·6°E, 35°S (dp = 2·5°, dm = 5·0°). This is linked to the K-Ar age to define a palaeopole of Lower-Upper

TABLE 1

Palaeomagnetic results from the E–W dyke swarm, Argyllshire, Northern Highlands of Scotland

Site no.	N	R	D (°)	I (°)	J_0 $\times 10^{-3}$ A . m^2/kg	Curie point(s) T_c (± 10°C)	Grid reference
1*	4	3·84	183	−2	0·01 – 0·04	—	2021 7811
2*	6	5·99	182	−5	20·00 – 34·0	530	1958 7793
3*	5	4·96	179	0	0·01 – 2·40	—, 500	1958 7793
4*	7	6·98	181	−10	10·7 – 17·5	580	1936 7799
5	5	4·95	199	−26	8·74 – 15·6	580	1856 7814
6	6	5·93	203	−23	17·00 – 87·30	570	1806 7827
7*	5	4·99	182	−7	1·62 – 9·31	580	1725 7844
8*	7	6·50	188	−20	6·96 – 12·50	350, 680	1727 7844
9*	6	5·86	169	7	46·70 – 66·30	550	2134 7906
10	4	3·91	325	64	13·70 – 23·10	620	2085 7917
11	7	6·94	181	−53	32·40 – 135·00	570	2023 7922
12	6	5·91	179	−57	2·80 – 96·40	580	2018 7923
13*	4	3·86	186	0	7·74 – 24·80	570	2142 7821
14*	5	4·93	178	−4	1·93 – 7·42	580	2148 7818
15*	5	4·99	182	−19	0·43 – 3·06	550	2149 7818
16*	6	5·92	177	−7	4·80 – 17·40	570	2176 7842
17*	5	4·97	174	12	0·67 – 11·28	570	2187 7863
18*	6	5·75	176	10	7·04 – 22·18	570	2188 7866
19*	5	4·93	180	−8	1·96 – 26·90	570	2185 7861
20*	5	4·86	187	−5	2·92 – 14·20	530	2131 7795
21	6	5·89	184	27	6·08 – 12·00	575	2145 7899

Mean of 15 sites (*)	R	D	I	k	α_{95}	Palaeomagnetic Pole °N	°E	dp	dm
	14·77	180·1	−3·8	60·9	4·9	−35·0	354·6	2·5	5·0

Footnote: D and I are the group mean declination and inclination of N samples with a resultant vector R and Fisher precision parameter k(=(N − 1)/(N − R)). Dp and dm are the radii of the oval of confidence about the mean palaeomagnetic pole in the colatitude direction and at right angles to it, respectively.

Carboniferous boundary age. Site 21 has a positive inclination over 20° steeper than the remaining dykes and nearer to Caledonian dolerites in this area (Esang and Piper 1984); for this reason we believe that it may belong to the older swarm and have excluded it from the group mean.

Discussion

Devonian and Lower Carboniferous results from Europe and North America have been collectively interpreted in terms of large scale tectonic displacement along the Great Glen Fault by Van der Voo and Scotese (1982). This model cannot

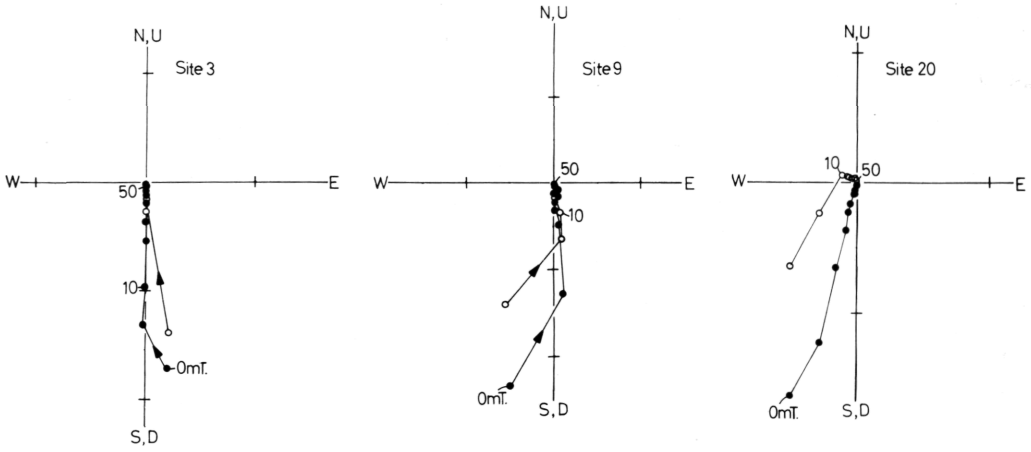


FIG. 2. Vector plots (Zidjerveld 1967) illustrating some typical a.f. demagnetisation behaviour of samples from the dyke swarm. Closed symbols represent projections on the horizontal plane, and the open symbols are projections on the vertical plane. The scale divisions are in 10^{-3} A m^2 kg^{-1}.

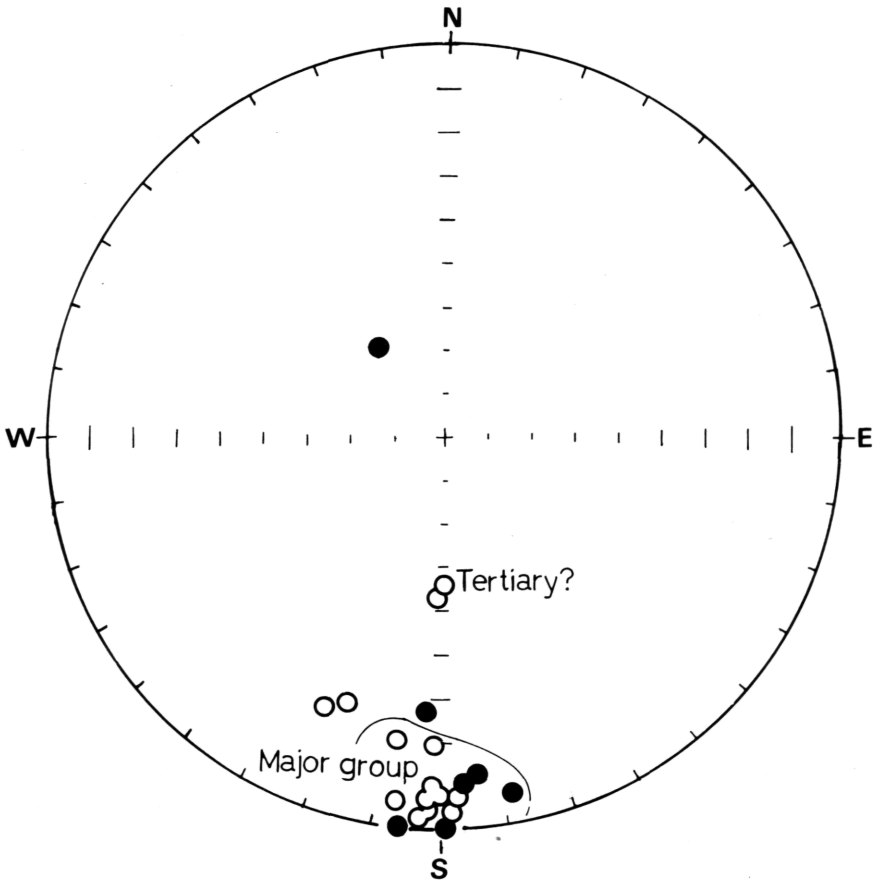


FIG. 3. Equal area projection of the site mean directions from the dyke swarm. Closed symbols are lower hemisphere plots and open symbols are upper hemisphere plots.

be reconciled with the continuity of Middle-Upper Old Red Sandstone sedimentation across the fault zone (Mykura 1975, Donovan *et al.* 1976), and correlations between the Moinian stratigraphy and metamorphic zonation across the fault zone (Winchester 1982). Furthermore, most recent analyses identify the complexity of the palaeomagnetic record and are unable to recognise a palaeomagnetically-detectable motion of Lower-Upper Carboniferous boundary age along the Great Glen Fault zone (Briden *et al.* 1984, Esang and Piper 1984). For these reasons, it is valid to compare the palaeopole from this study with data from Britain south of the fault zone and the remainder of continental Europe.

The expected path of the south palaeomagnetic field direction in the Northern Highlands derived from these data is shown in Figure 4. The mean direction of magnetisation of the E–W dykes coincides precisely with this path at 320 Ma in

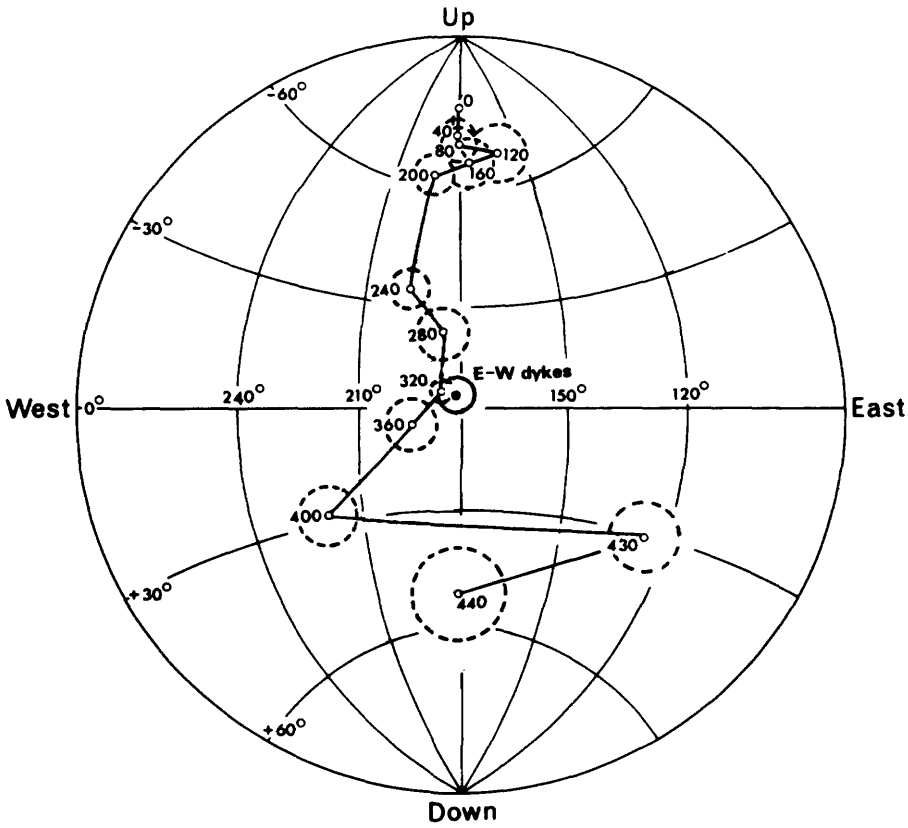


FIG. 4. The palaeomagnetic vector derived from the E–W dyke swarm compared with the reversed geomagnetic field direction in the Northern Highlands of Scotland from Ordovician times to the present. The expected path of the ancient field is derived in part from other British data and in part from global palaeomagnetic data and is modified after Briden and Mullen (1984).

accord with the K-Ar determinations. It is significantly removed from the path at 280–240 Ma and thus does not support a Permo-Carboniferous age.

Acknowledgements

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